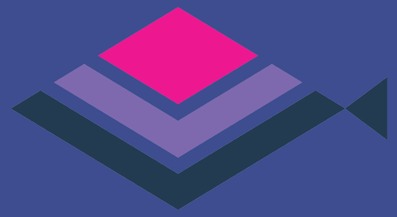

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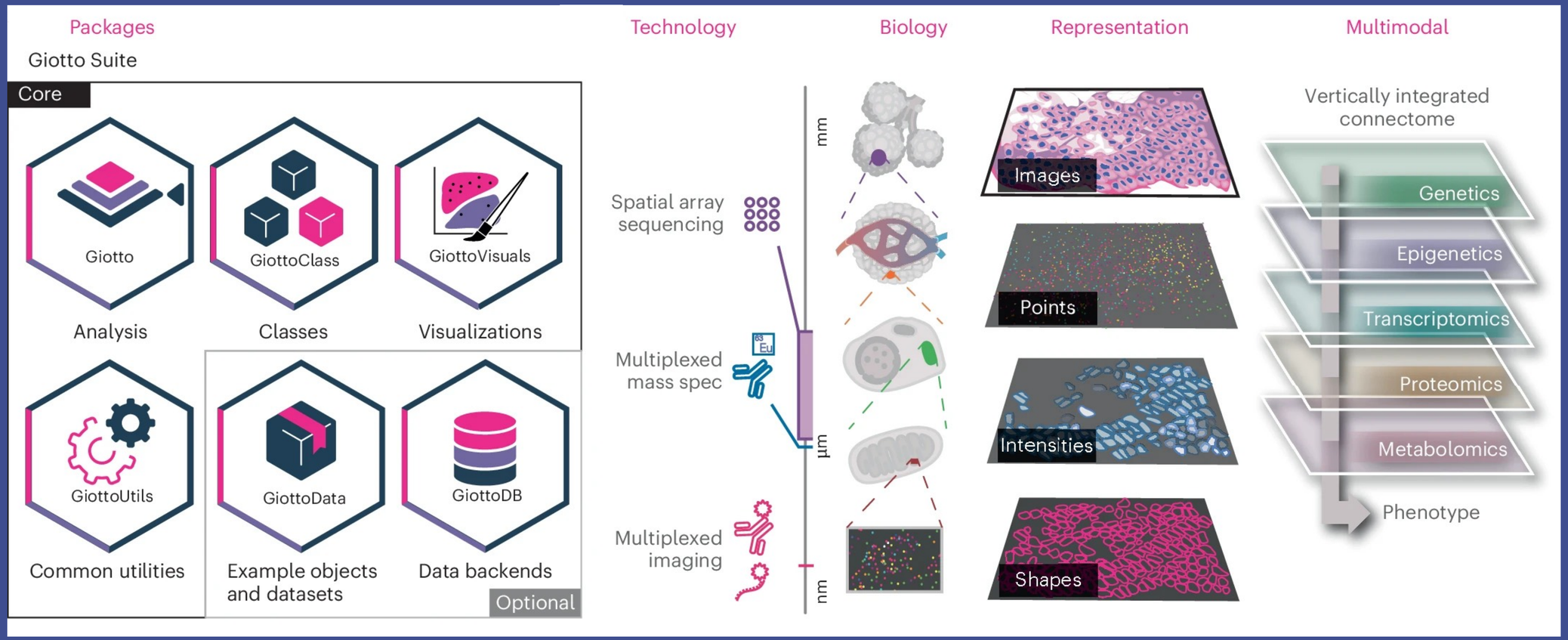
Spatial multiscale and multiomics integration with Giotto Suite

Giotto Suite is a technology agnostic, multi scale and multimodal tool, organized as a group of modular packages



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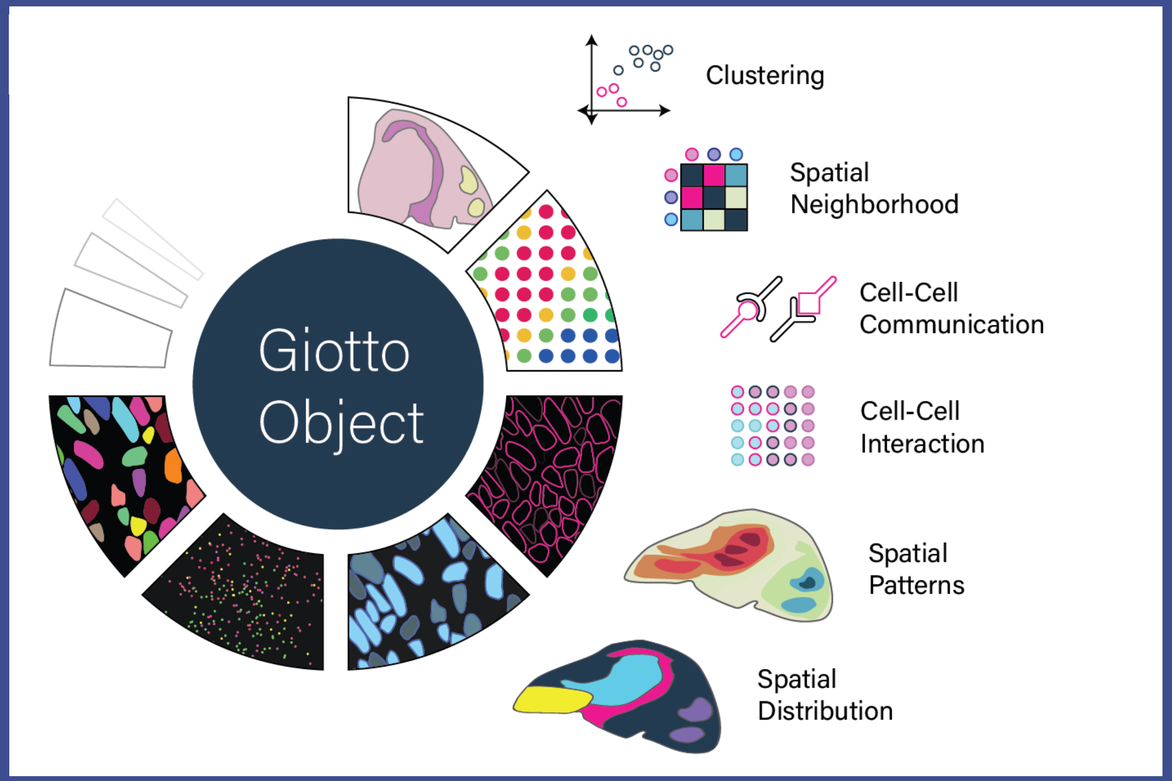
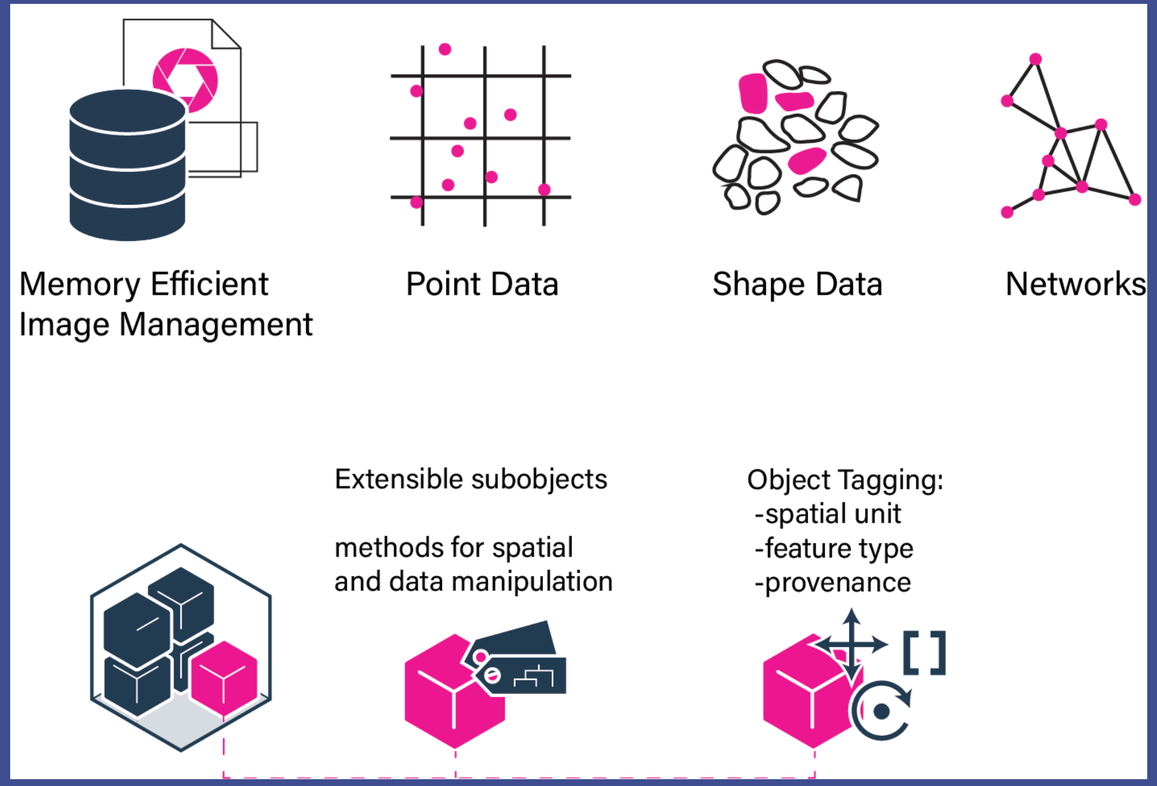
Giotto Suite provides scalable and extensible end-to-end solutions





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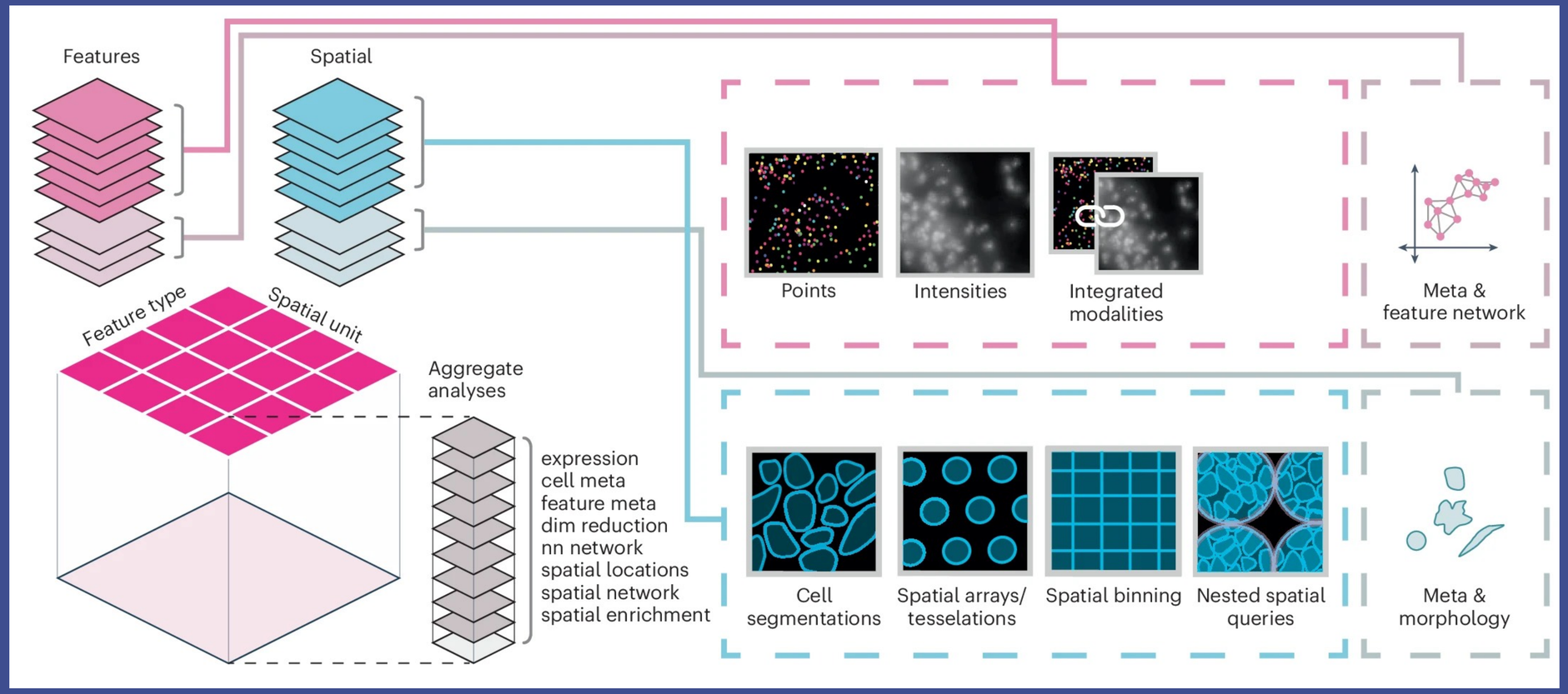
Giotto Suite's core framework facilitates joint data representation and analysis at any level



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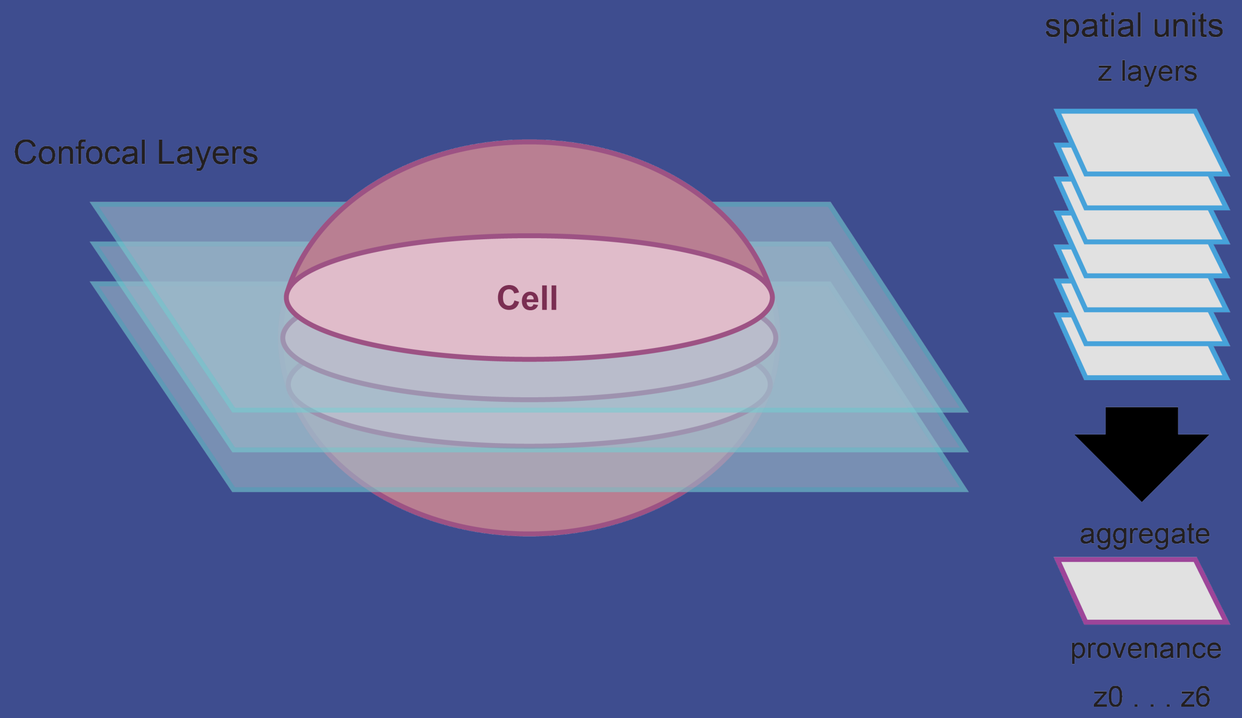
The multi spatial unit framework facilitates 3D transcript detection and aggregation across z stacks



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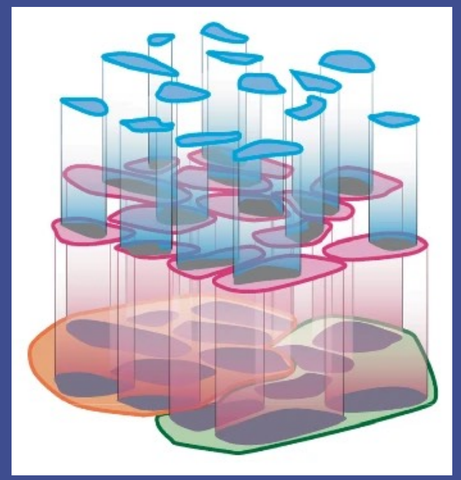
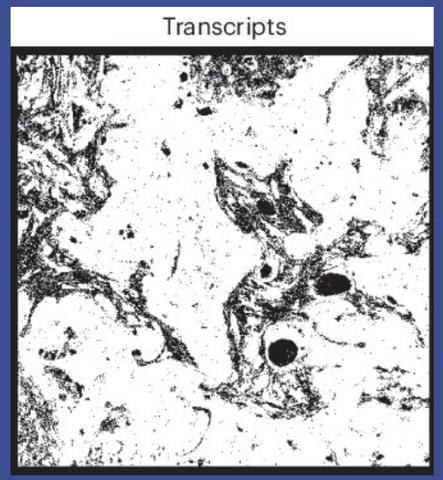
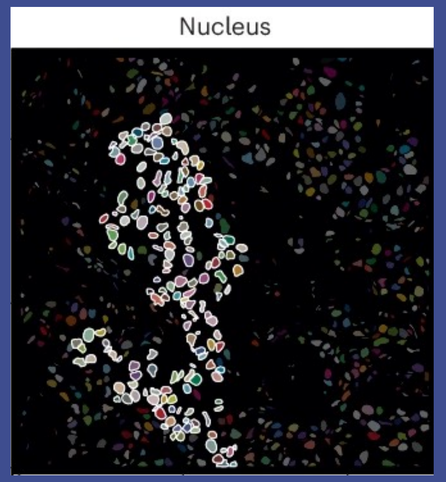
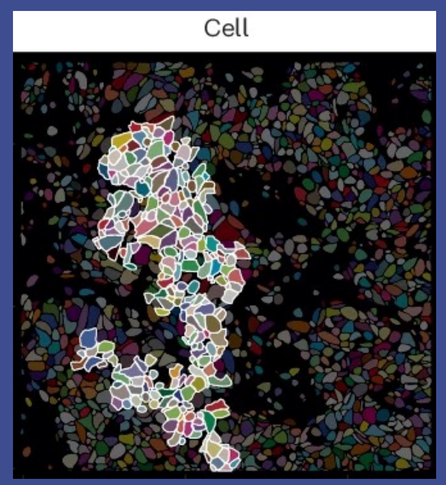
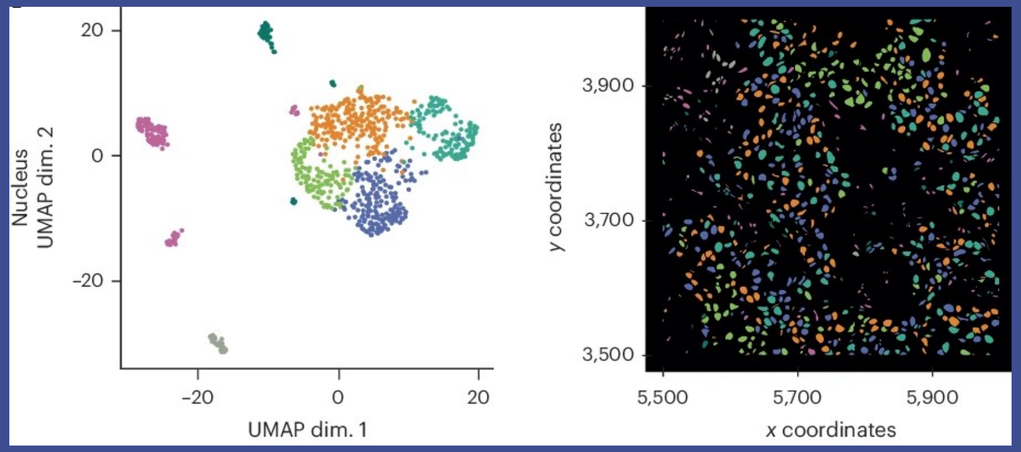
Transcript locations can be queried against predefined spatial units and used to identify genes that are spatially enriched at subcellular organelles

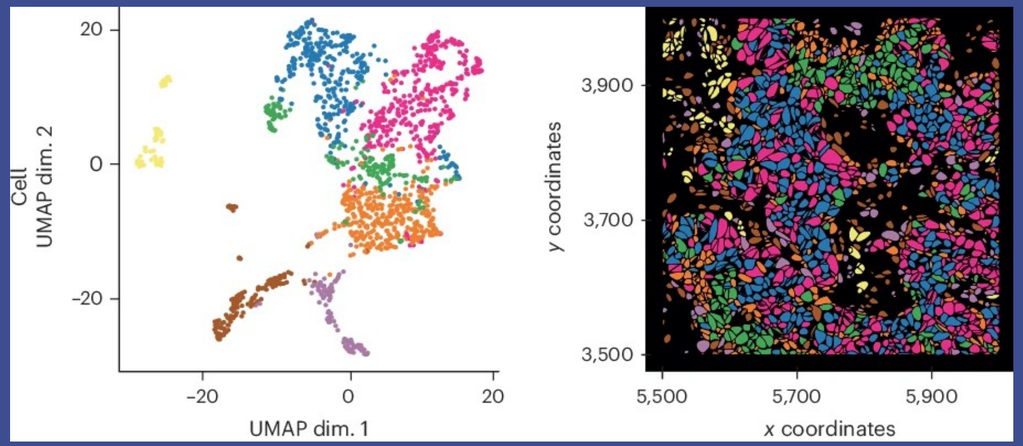












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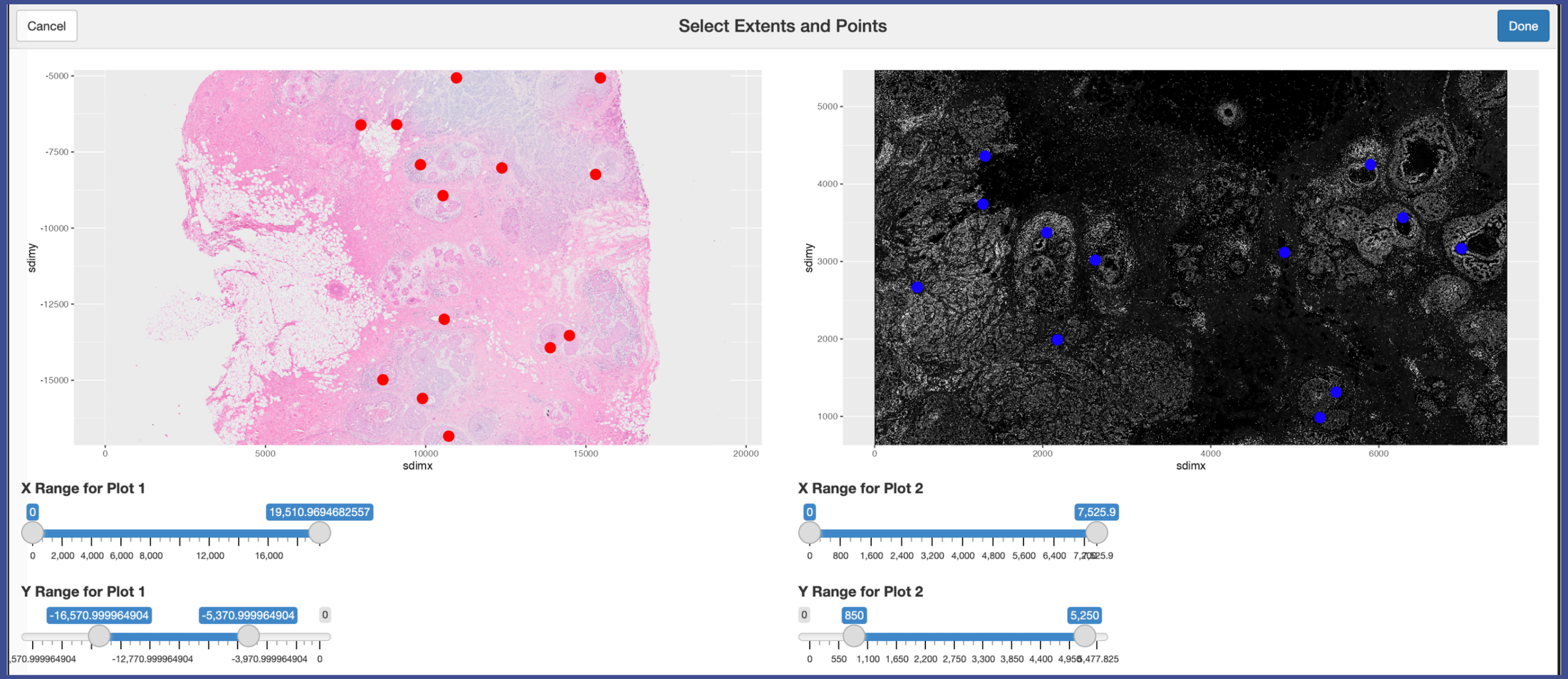
Giotto contains and interactive tool for multimodal landmarks selection and affine transformation



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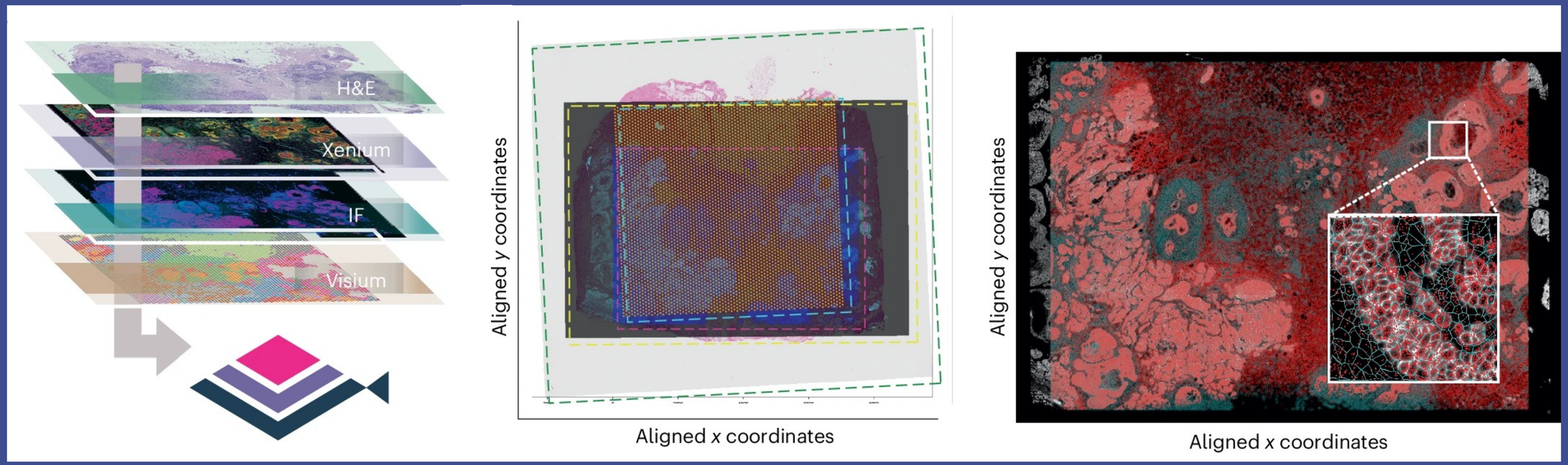
The transformed spatial datasets are used to create multimodal co-registration



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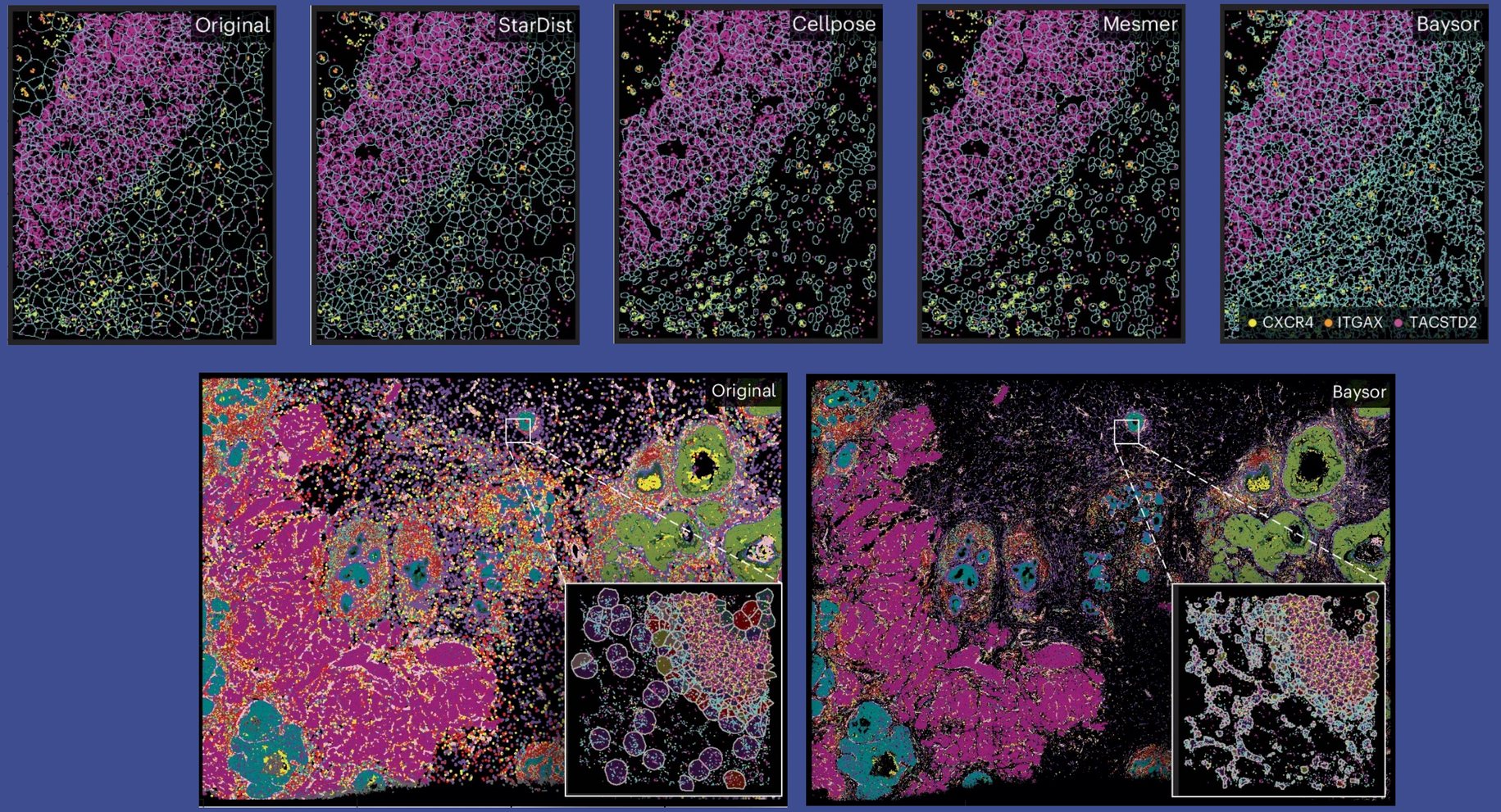
Giotto Suite can read and simultaneously store various segmentation output formats, including mask files or geojson files



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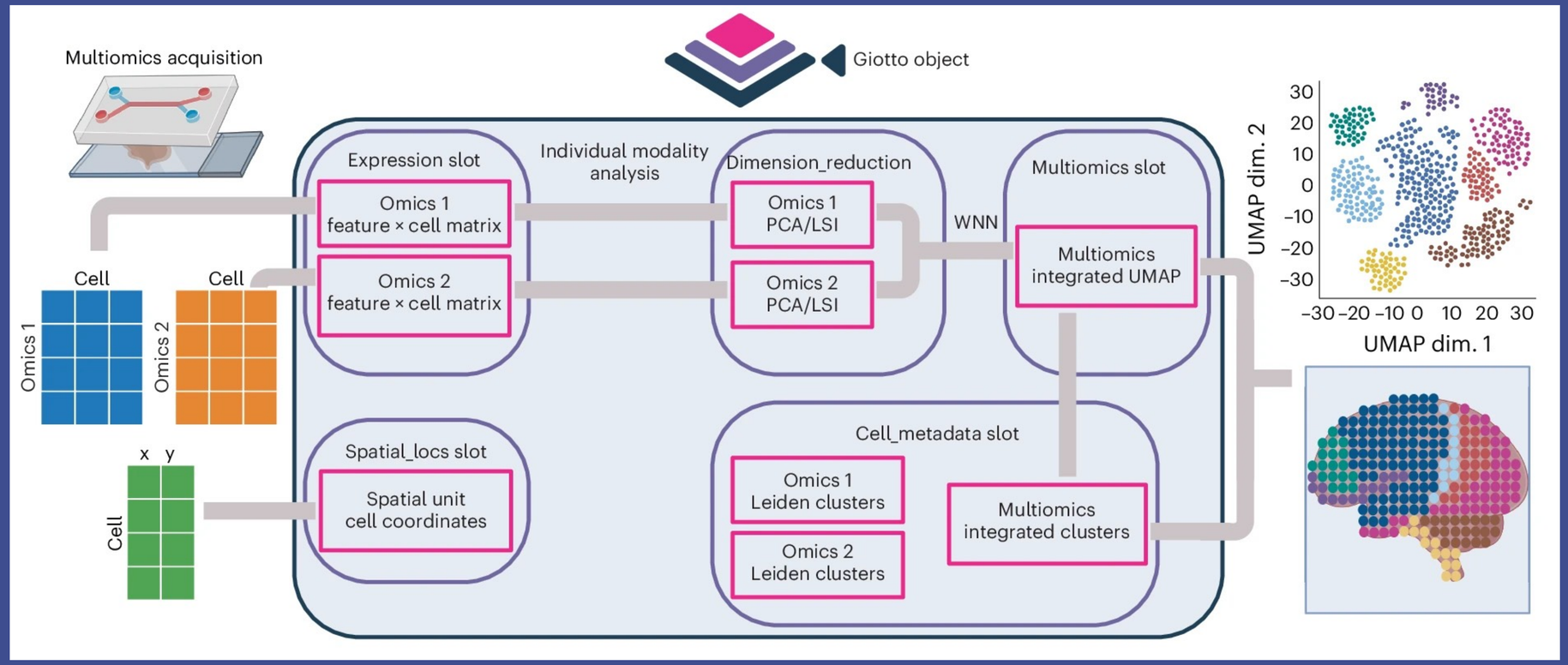
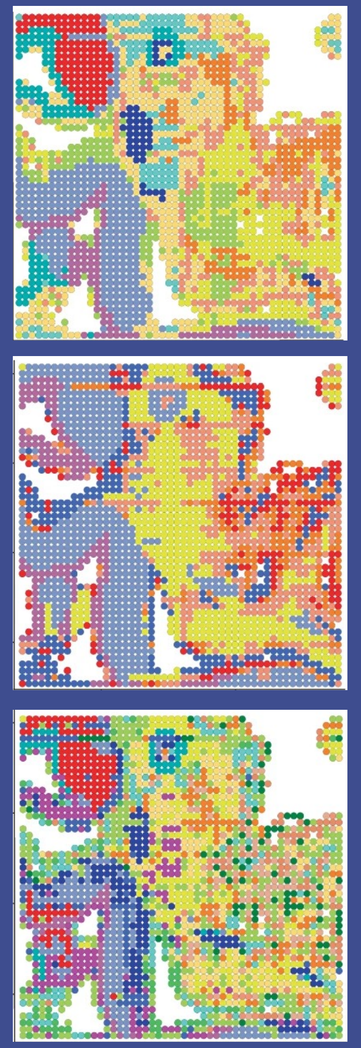
Giotto Suite supports multi-omics integration to obtain a more comprehensive characterization of cell states





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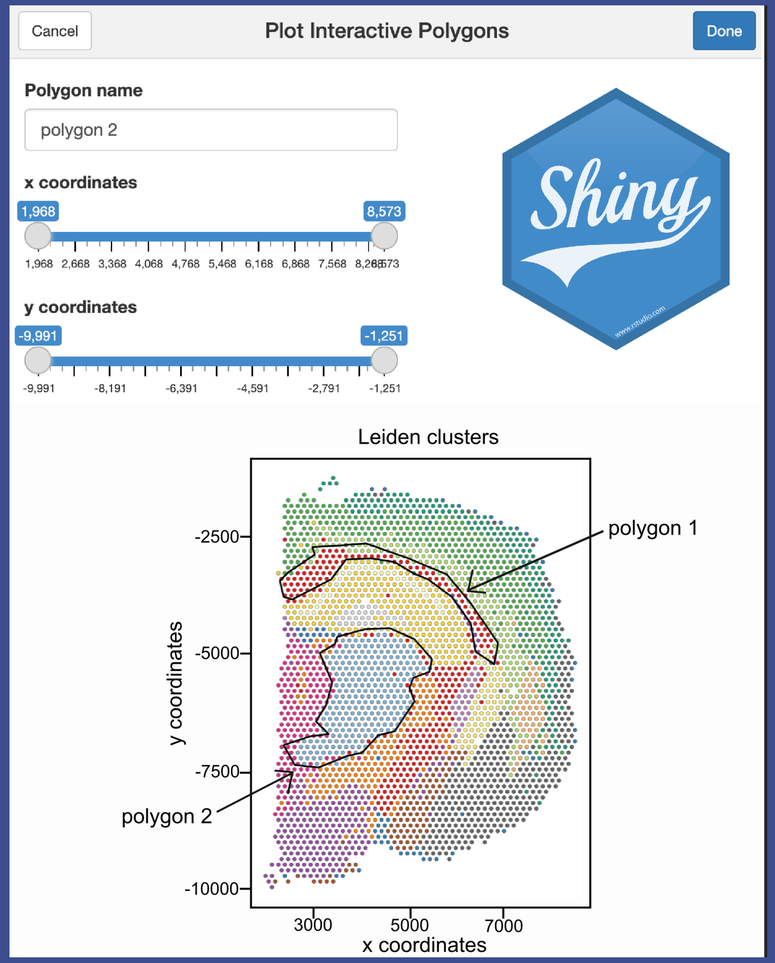
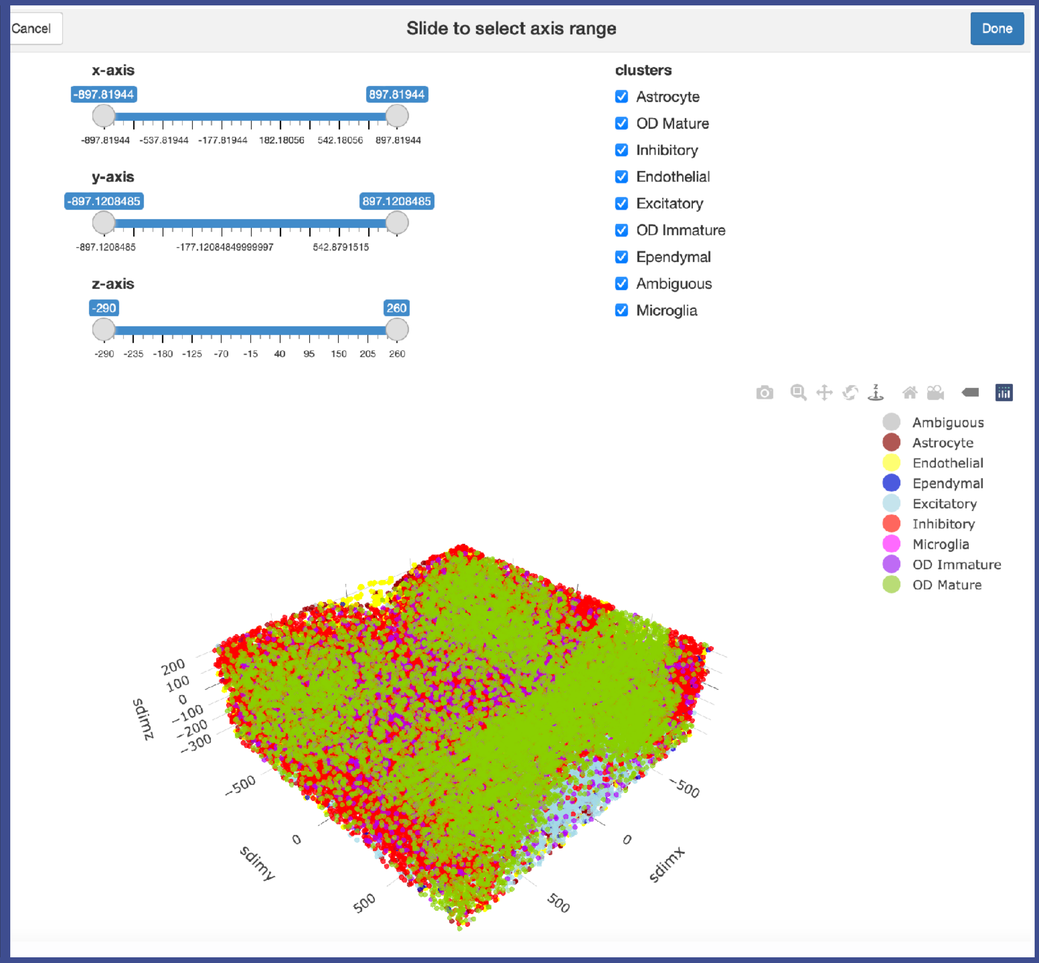
Giotto provides a 2D and 3D interactive selection tool compatible with downstream analysis

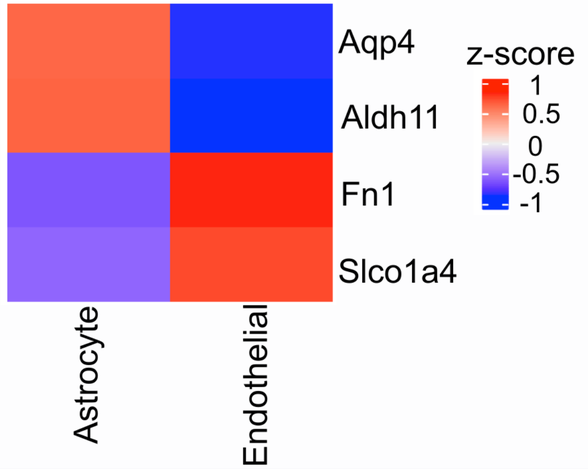




Cells subsetting

```
> my_slice_coordinates
        cell_ID     sdimx     sdimy sdimz  cell_types
   1: cell_17230 -892.5276 -848.3141   110   Astrocyte
   2: cell_17233 -885.8688 -803.6590   110 Endothelial
   3: cell_17235 -877.4924 -768.5620   110 Endothelial
   4: cell_17244 -849.7980 -752.6687   110   Astrocyte
   5: cell_17246 -847.2183 -885.0670   110 Endothelial
  ---
1377: cell_23016  712.5530  305.5107   110 Endothelial
1378: cell_23021  821.0644  494.4085   110 Endothelial
1379: cell_23026  893.1285  837.7114   110   Astrocyte
1380: cell_23027  895.7814  873.2376   110 Endothelial
1381: cell_23028  895.2894  825.7128   110 Endothelial
```

Gene expression patterns



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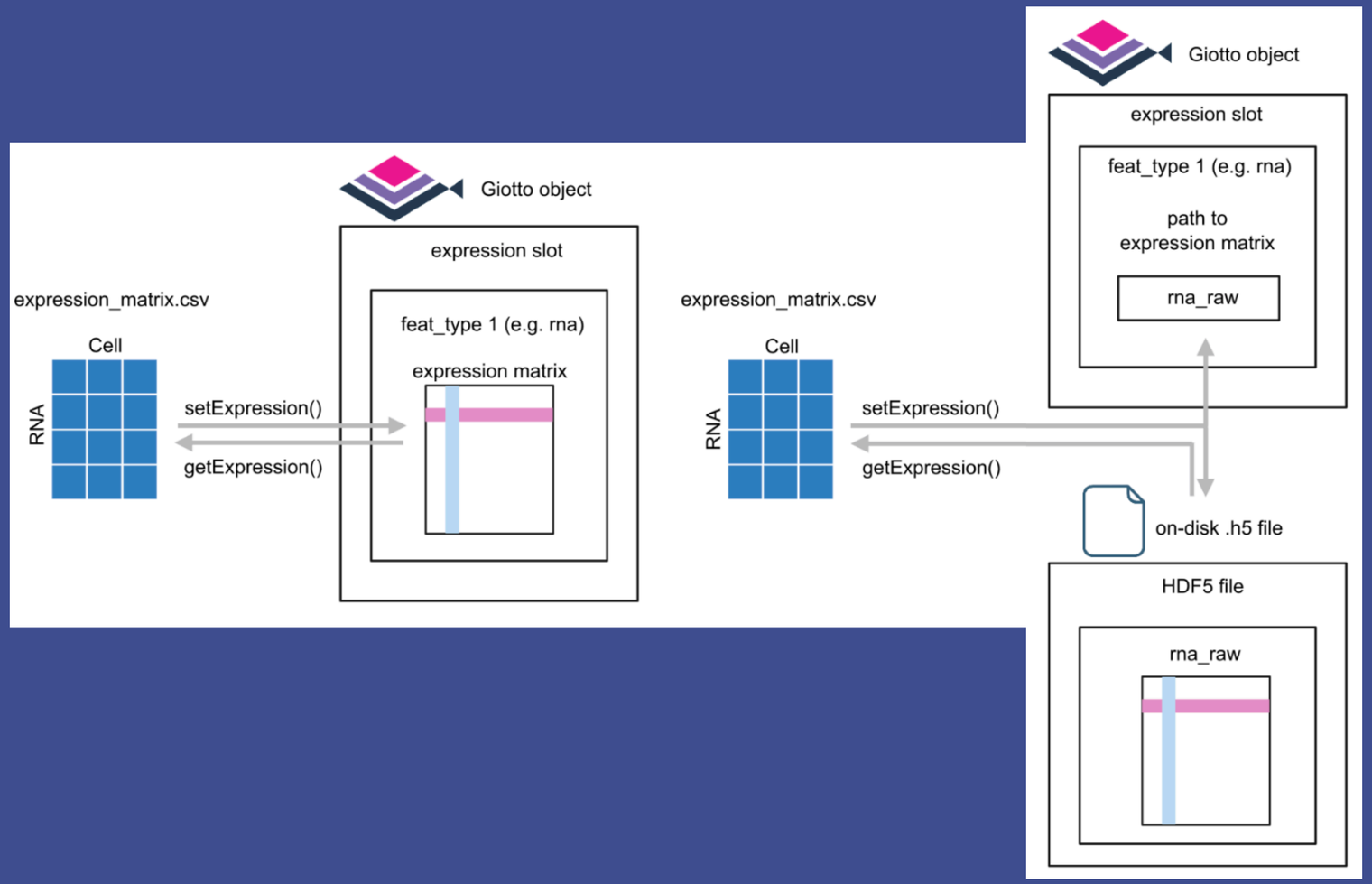
Giotto implements several complimentary tools for scalable data analysis (delayed on-disk calculations and database backend)



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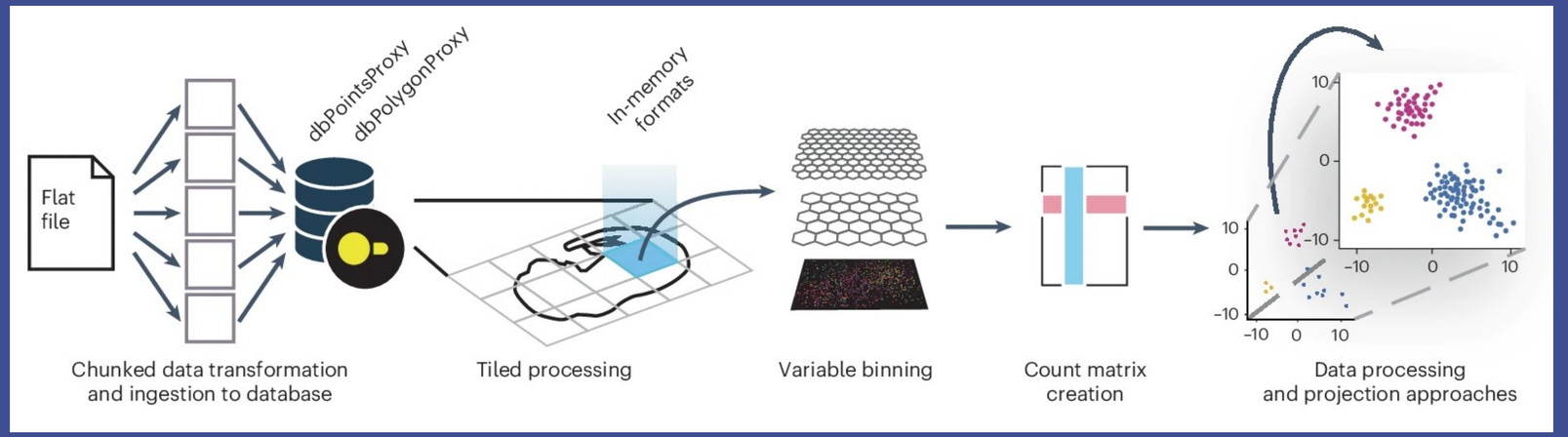
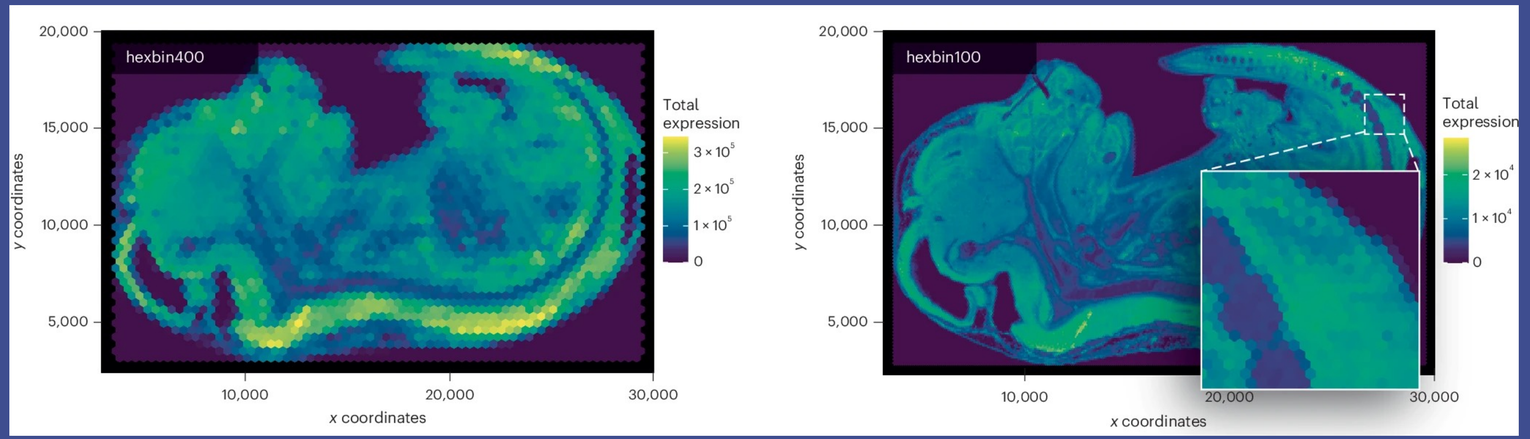
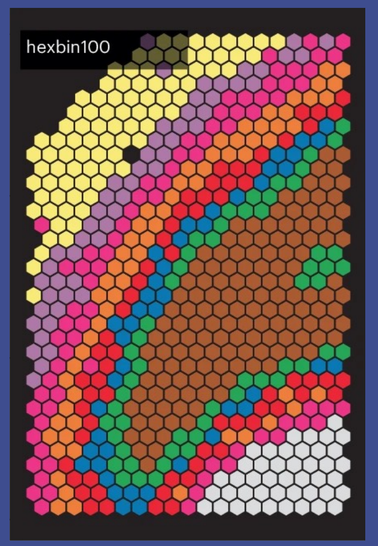
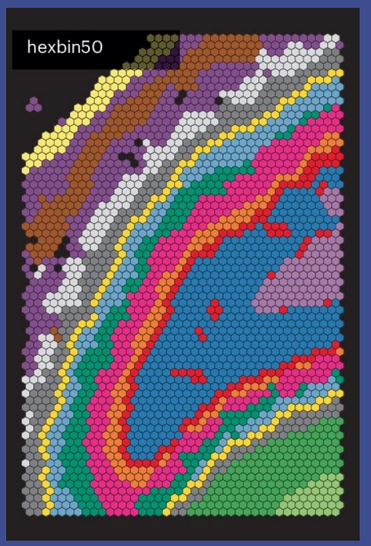
Giotto Suite also offers flexible tiling and tessellations to analyze large-scale data at multiple scales or resolutions









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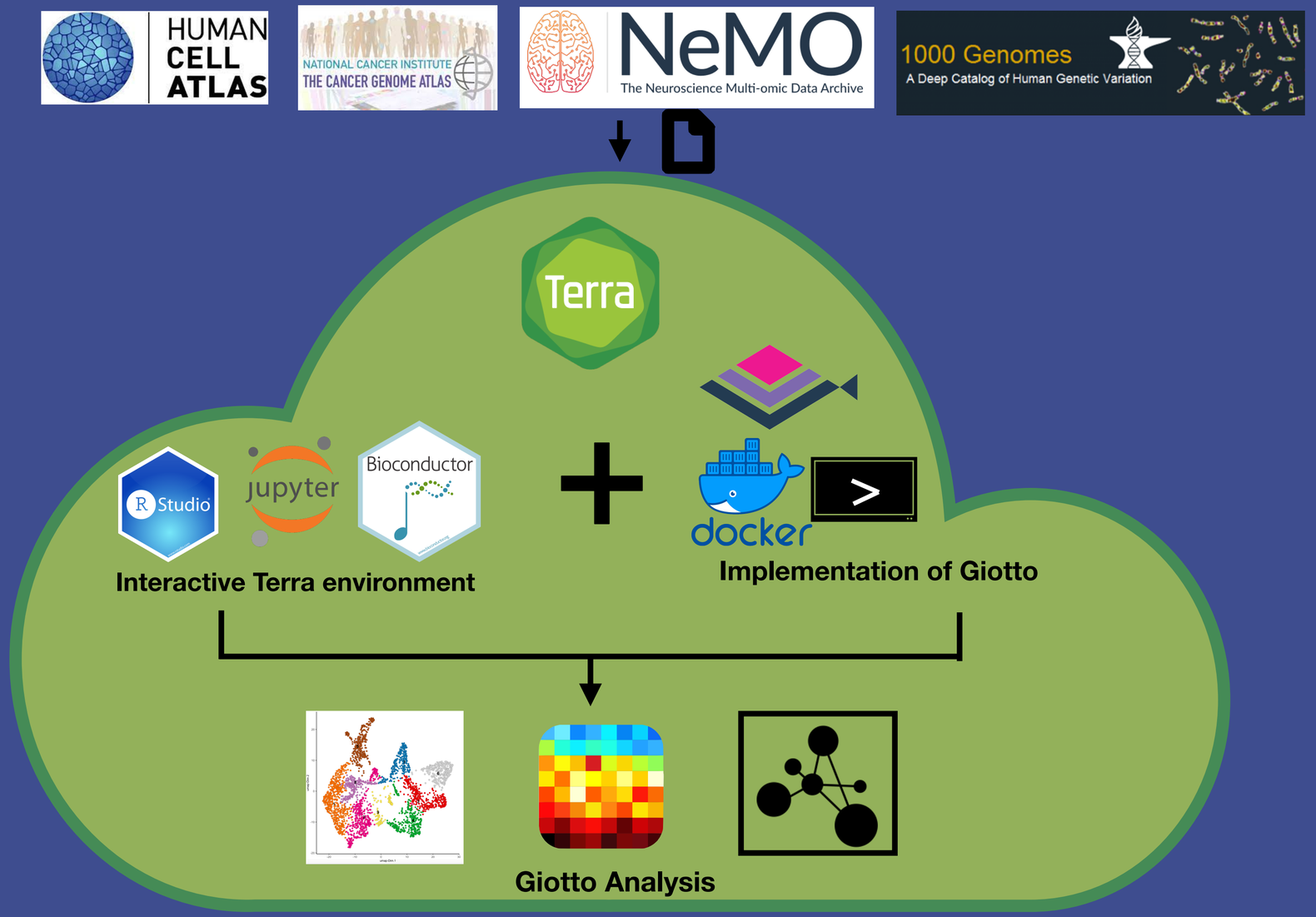
Giotto Suite is also built, tested and available on terra.bio



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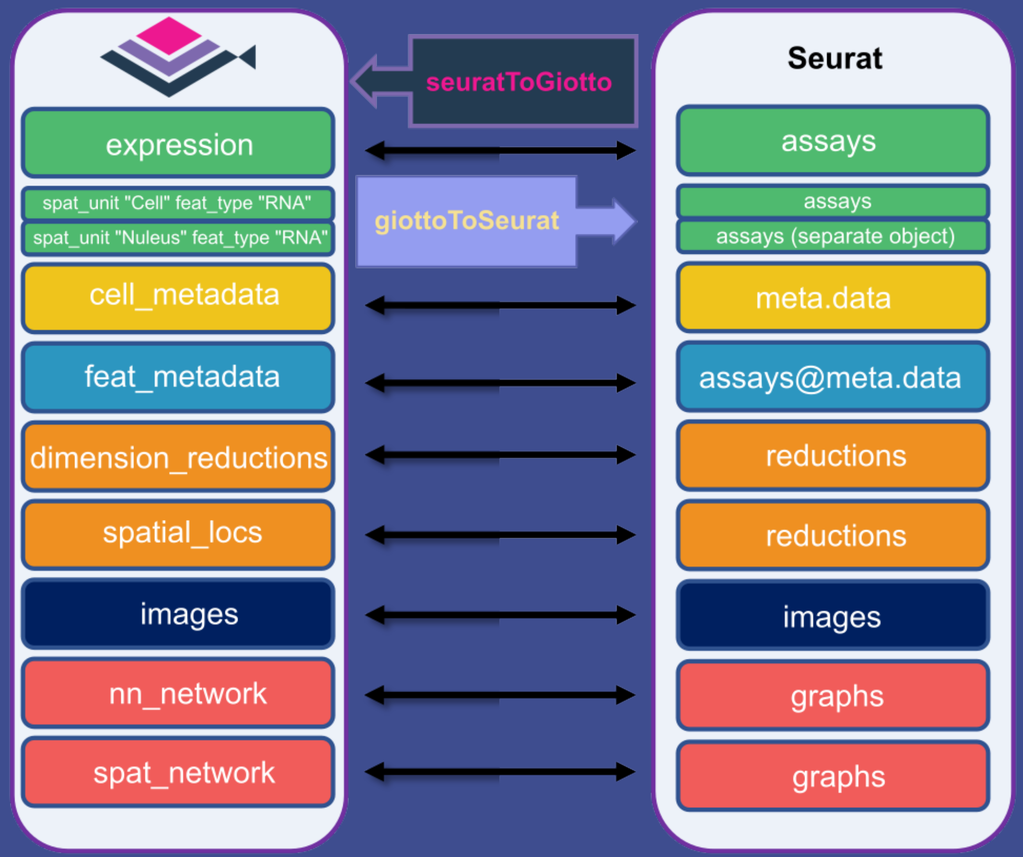
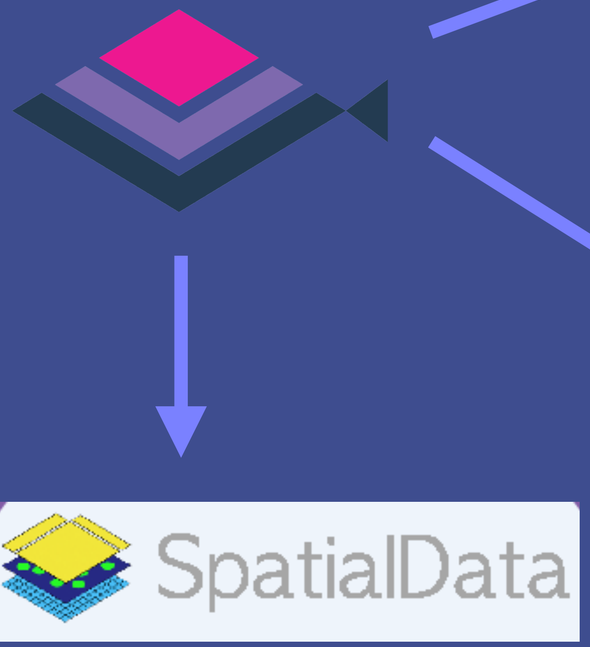
Giotto Suite provides several utilities to facilitate interoperability with external tools





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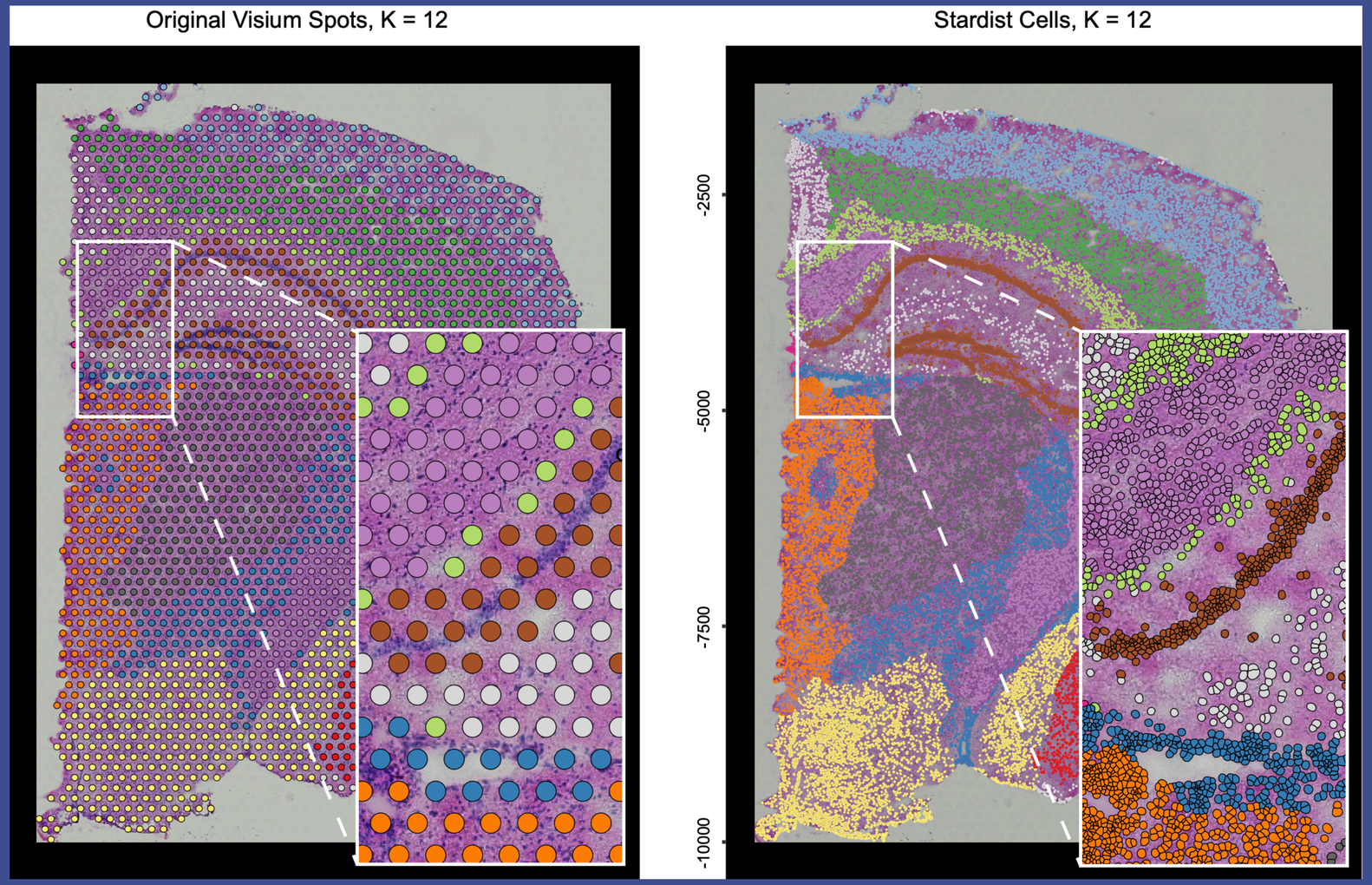
Interpolation methods, such as ‘kriging’ in the ‘gstat’ package, can be easily combined with Giotto to enhance low-resolution spatial datasets



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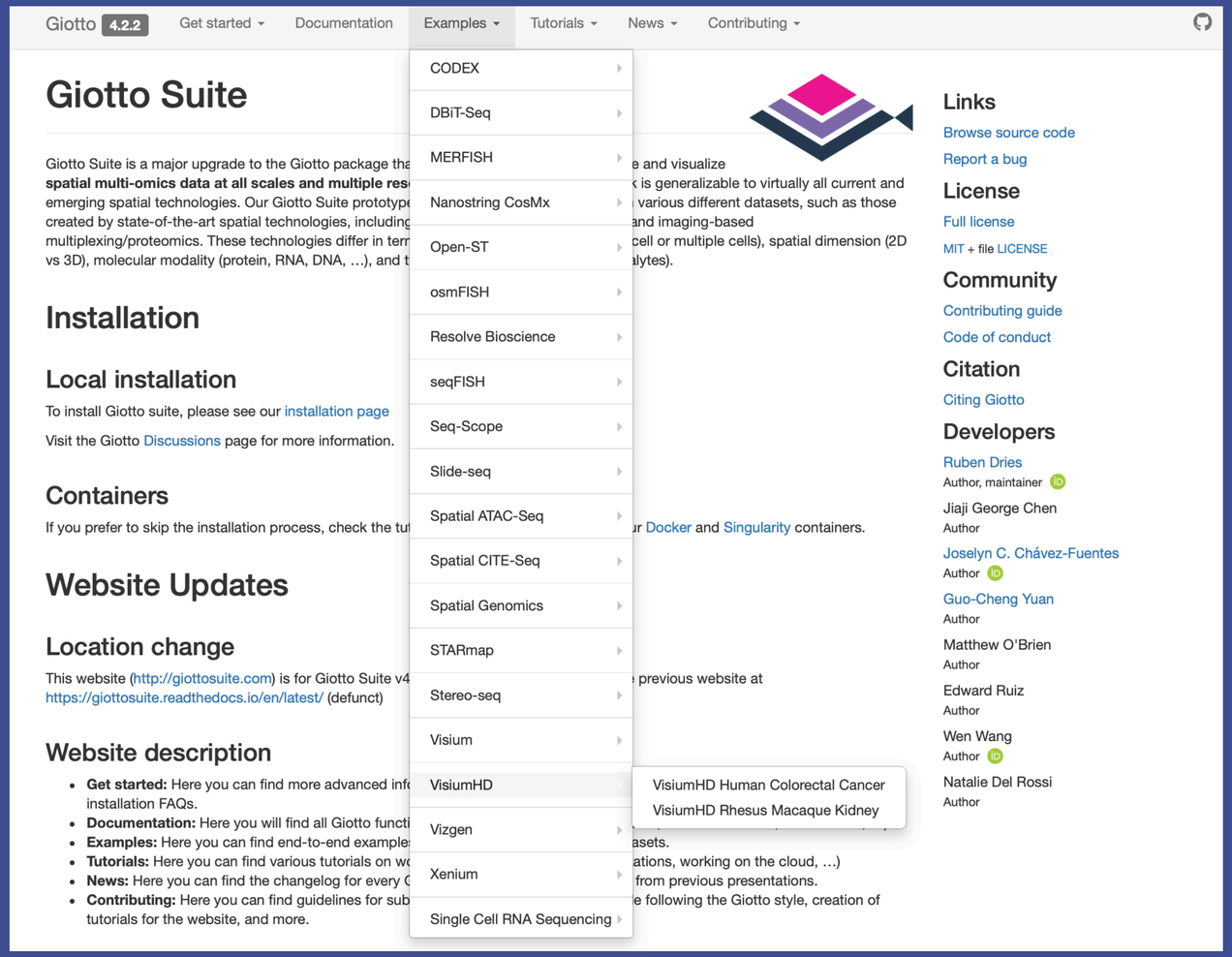
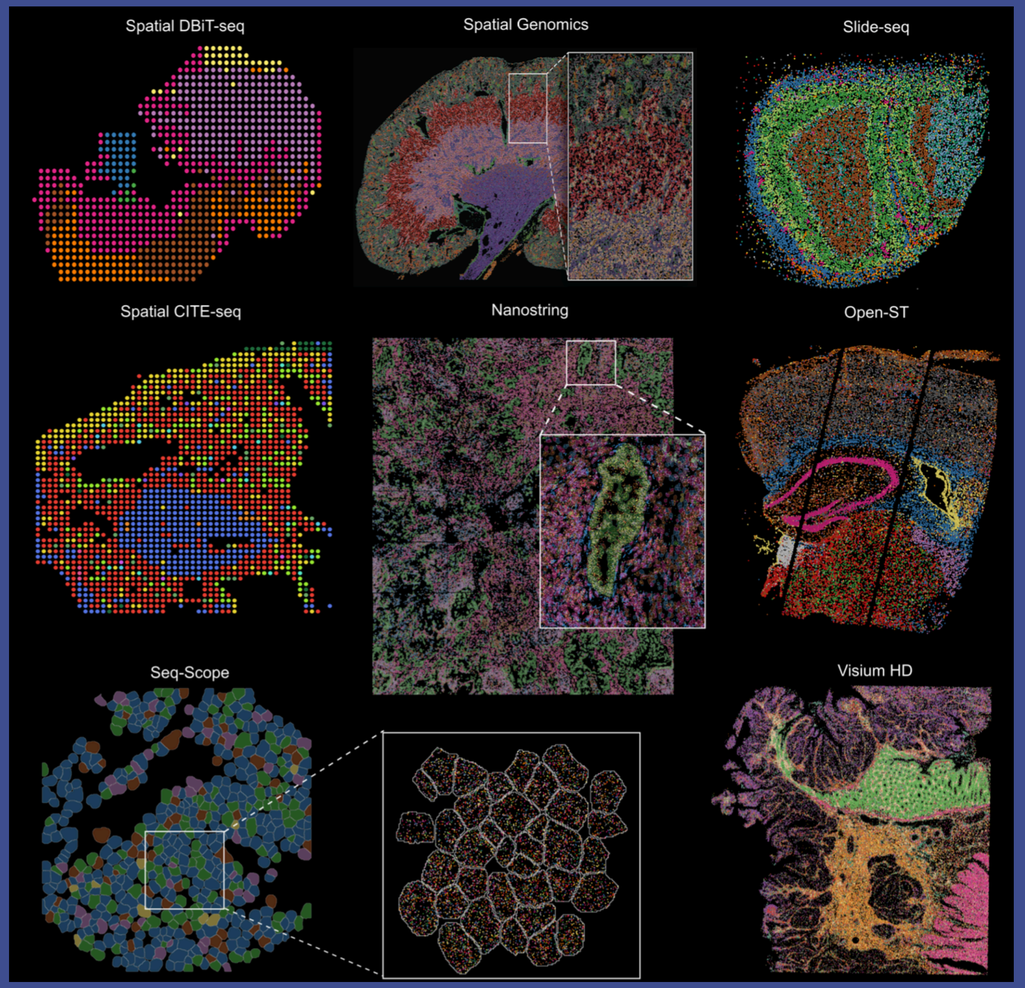
General and technology-specific tutorials are available on the Giotto Suite website





http://giottosuite.com

Acknowledgments

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NIH National Institute of Mental Health

Chan Zuckerberg Initiative